Ordering Fractions, Decimals & Percentages

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Ordering FDP

How do I put fractions in order of size?

- When comparing **only fractions**, write them over a **lowest common denominator**
  - For $\frac{3}{5}, \frac{1}{2}, \frac{13}{20}, \frac{7}{12}$, the lowest common denominator is **60**
  - So change them to $\frac{36}{60}, \frac{30}{60}, \frac{39}{60}, \frac{35}{60}$ and then **order them by their numerators**
  - From smallest to largest: $\frac{30}{60}, \frac{35}{60}, \frac{36}{60}, \frac{39}{60}$
  - Rewrite in their original form: $\frac{1}{2}, \frac{7}{12}, \frac{3}{5}, \frac{13}{20}$

How do I put fractions, decimals and percentages in order of size?

- When comparing a mixture of fractions, decimals and percentages, **change everything into decimals**

e.g. Write the following in order of size, starting with the smallest. (do not use a calculator)

$0.31 \quad \frac{2}{3} \quad \sqrt{0.09} \quad 32\% \quad 3^{-1}$

Step 1: Convert to decimals...

0.31	Already a decimal
$\frac{2}{3} = 0.666\,666\ldots \ (0.\dot{6})$	Recognise
$\sqrt{0.09} = 0.3$	Recognise from $\sqrt{9} = 3$
$32\% = 0.32$	$32 \div 100$
$3^{-1} = \frac{1}{3} = 0.333\,333\ldots \ (0.\dot{3})$	Negative power means "1 over"

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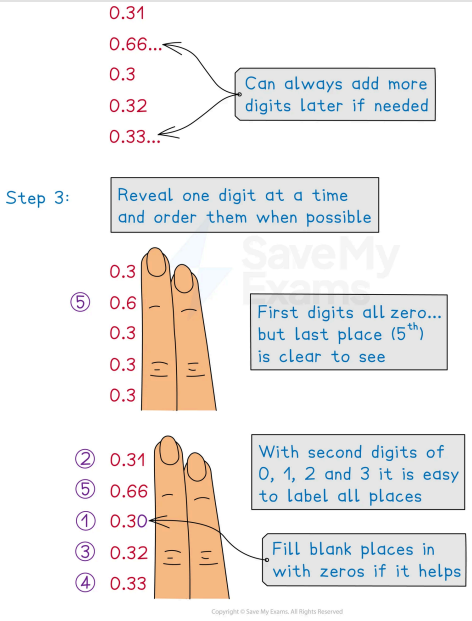

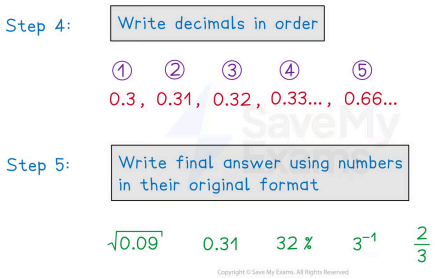

Which symbols can I use?

- Rather than just listing values in order, symbols can be used to compare them
  - For example, $\frac{1}{4} < \frac{1}{3} < \frac{1}{2}$
- Recall that $>$ means **greater than** and $\geq$ means greater than **or equal to**
  - Similarly, $<$ means **less** than and $\leq$ means less than or equal to
- You may also see $=$ and $\neq$ (which means "**not** equal to")

Examiner Tips and Tricks

A calculator can be used to quickly convert any quantities into decimals.

E.g. A fraction can be entered as a division.

Without use of a calculator, write these numbers in order, starting with the smallest.

$$\frac{1}{3}, \frac{2}{5}, \frac{9}{25}, \frac{4}{15}$$

Answer:

As they are all fractions, write them with a common denominator
The lowest common denominator is 75
Rewrite each fraction with a denominator of 75

$$\frac{1}{3} = \frac{25}{75} \quad \frac{2}{5} = \frac{30}{75} \quad \frac{9}{25} = \frac{27}{75} \quad \frac{4}{15} = \frac{20}{75}$$

Compare and write in order from smallest to largest

$$\frac{20}{75}, \frac{25}{75}, \frac{27}{75}, \frac{30}{75}$$

Rewrite in their original form

$$\frac{4}{15}, \frac{1}{3}, \frac{9}{25}, \frac{2}{5}$$

Worked Example

Without use of a calculator, write these numbers in the spaces below.

$$\frac{7}{8}, \frac{5}{6}, 0.8, 78\%$$

..... < ..... < ..... < .....

Answer:

As there is a mixture of fractions, decimals, and percentages, rewrite each as a decimal

0.8 is already a decimal

Convert 78% to a decimal by dividing by 100

$$78\% = 0.78$$

Convert $\frac{7}{8}$ to a decimal by either using short division, or halving three times

$$7 \div 2 = 3.5$$
$$3.5 \div 2 = 1.75$$
$$1.75 \div 2 = 0.875$$
$$\text{so } \frac{7}{8} = 0.875$$

Convert $\frac{5}{6}$ to a decimal by using short division

$$\begin{array}{r} 0.833\ldots \\ 6\overline{)5.000\ldots} \\ {\scriptstyle 5\,2\,2} \end{array}$$

Write them all with 3 decimal places to determine the order

$$0.875$$
$$0.833$$
$$0.800$$
$$0.780$$

Rewrite the decimals in order

$$0.78,\ 0.8,\ 0.8\dot{3},\ 0.875$$

Rewrite in their original form, and recall that < means "less than", so the smallest value will be first

$$78\% < 0.8 < \frac{5}{6} < \frac{7}{8}$$

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